GRAVITYSORT · STEM CHECK

GravitySort STEM check

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gravitysort		
Source	docs/01b-stem-check.md @ 6c7e5bc		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.8 pipeline on a TRL 3 design): 40 claims for options A, B and C; five details failed and were fixed under Amish's standing instruction of 2026-10-23 (GVS-DEC-001 F1 to F4, F6); two rows fail with fixes that go to Amish (S10, S33); independent review recorded

GravitySort STEM check

Verdict: no claim fails outright. Of 40 claims, 29 sit within known science after five fixes made under Amish's standing instruction of 2026-10-23; 9 are plausible but need a test or a datasheet at TRL 4 (fine-gold retention in the bed, critical speed on the real frame, table ratio, the motor's speed cap, filter performance, liner casting and wear, borax smelting of a sulfide-rich concentrate, option C's hand panning); and 2 fail with a fix proposed that goes to Amish because each fix breaks the 100 kg of PR-011 (containment of a bowl fragment, S10; the full tank's sway near the table drive frequency, S33). So `stem_check` is **open**. Method and rules: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py` (writes `stem_results.csv`), which imports the calculation note GVS-CAL-001 (`sizing.py`) so that every row on the TRL 3 design closes against the same numbers.

Fixes made under the standing instruction (GVS-DEC-001)

This repo was already at TRL 3 when its first STEM check ran (the kit 1.8 upgrade), so it keeps its TRL and the fix rule applies to the failed items. Before the fixes these five rows were **Fails, fix proposed**; each fix keeps the working principle, the pitch and the scope, so it was made in the same session and the row closed (S4 was found by the independent review):

Row	What failed	Fix (GVS-DEC-001)
S20	The header drum was drawn 380 mm across and 400 mm tall: 45 L, not the 60 L used everywhere else	F1: drum drawn at its real 560 mm height; head and tipping rechecked (S4, S11)
S24	Recirculated, silty pond water went unfiltered to 1.0 mm fluidization holes; fluidized concentrators need clean fluidization water	F2: 3/4 in screen filter, 120 mesh, with a flush valve; supply rechecked with its loss (S4, S5)
S16	The BOM's buffer rating (600 N) was barely above the peak force at the largest setting (594 N)	F3: buffer rated 800 N or more, as GVS-CAL-001 section 9 had chosen
S21	The gold balance used 8 h of feed although the restated throughput has 7.75 h	F4: gold balance on the feed time (7.75 g in per shift)
S4	The supply calculation left out the climb from the union to the rings (4.5 to 5.6 kPa); with it the lowest ring had 6.0 kPa net at 600 rpm against the proposed 10 kPa	F6: ball valve and rotameter with 3/4 in bores (fittings loss 7.5 to 3.4 kPa); 10.0 kPa at mid-tank, 7.0 kPa with the tank nearly empty and the filter due for flushing

The detailed CAD of the kit 1.9.1 upgrade found one more clash, not a science row: the head pillow block nuts hit a shelf gusset, fixed by moving the gusset (F5). Two rows fail with fixes that break PR-011 and therefore go to Amish: S10 (containment, GVS-DEC-001 O2) and S33 (tank sway, O3).

Claims checked

Table 1. Claims, governing science and verdicts (from `docs/04-calcs/stem_results.csv`).

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	The bowl gives about 60 G at the 100 mm reference radius at 730 rpm	Physics	Mechanics and statics	Centripetal acceleration a = omega^2 r	claimed 60 G, estimate 59.6 G (ratio 1.01)	Within known science	GVS-CAL-001 section 2
S2	Fine gold reaches the bowl wall before it leaves: a 20 um gold flake crosses the 5 mm film well within the transit time	Physics	Fluid mechanics	Terminal velocity in the centrifugal field with Schiller and Naumann drag; flake at half the sphere velocity	36 transit/ settle ratio >= 1 transit/settle ratio	Within known science	GVS-CAL-001 section 5
S3	The fluidized bed in the riffle rings keeps fine gold (80 % of 75 to 1,000 um, 50 % of 38 to 75 um; about 71 % of screened gold) while the bed is flushed every 2 h	Physics	Fluid mechanics	Fluidized centrifugal concentrators recover very fine gold over a wide size range (Veiga and Gunson 2020). The 2.7 mm/s upflow is about the minimum fluidization velocity of 75 um quartz at 60 G (2.9 mm/s, Wen and Yu) and well below that of coarser or denser grains (27 mm/s for 150 um pyrite), so most of the bed is not fully fluidized; commercial bowls use 10 to 30 L/min at 35 to 100 kPa (iCON i150) against 12 L/min at about 10 kPa here; saturation by heavy minerals in the feed is not calculable	Settling check passes (S2); retention needs a test with characterized or spiked ore (PR-004, TRL 4)	**Plausible, to confirm at TRL 3**	Open: test or datasheet at TRL 4

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S4	The 1.7 m header post gives the proposed 10 kPa net at every ring at 600 rpm with the tank half full, without a pump	Physics	Fluid mechanics	Static head less hose, fittings and filter losses (Darcy-Weisbach, K factors), less the climb from the union to the ring (rho g dz), plus rho omega^2 r^2 / 2 in the jacket. v0.8 left out the climb (4.5 to 5.6 kPa) and had 6.0 kPa: fixed by 3/4 in valve and rotameter	claimed 10 kPa, estimate 9.96 kPa (ratio 1.00)	Within known science	GVS-DEC-001 F6; GVS-CAL-001 section 6
S5	Water still enters every ring with the tank nearly empty and the filter at its cleaning point	Physics	Fluid mechanics	Lowest net ring pressure at 600 rpm, tank 50 mm above its floor, filter K 6	6.96 kPa >= 0 kPa	Within known science	GVS-CAL-001 section 6; GVS-DEC-001 F2
S6	Bringing the slurry and water to lip speed takes about 27 W at 730 rpm	Physics	Thermodynamics	Power = m v^2 at the lip: half leaves as kinetic energy, half is lost in slip in the film (an upper bound); bowl and table never run together	claimed 27 W, estimate 27.2 W (ratio 0.99)	Within known science	GVS-CAL-001 section 7
S7	Bowl, jacket and rings keep a burst safety factor of at least 10 at 1,200 rpm (a sprinting rider)	Physics	Strength of materials	Thin-wall hoop stress from jacket water pressure and self-weight; GFRP 80 MPa, PU 20 MPa (typical)	16 safety factor >= 10 safety factor	Within known science	GVS-CAL-001 section 10
S8	The spindle's first critical speed is well above the highest reachable bowl speed (1.25 x 1,200 rpm)	Physics	Acoustics and vibration	Cantilever 3EI/L^3 of the 25 x 2 mm tube gives 2710 rpm on rigid bearings; with the compliance of the 25 x 25 x 1.5 members the bearings hang	Above 900 rpm, the limit, in every case; against 1,500 rpm only if the members act nearly fixed. Tap test on the first frame at	**Plausible, to confirm at TRL 3**	Open: test or datasheet at TRL 4

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				from, magnified by the 210 mm overhang over the 165 mm bearing span, 1274 rpm (member ends pinned) to 1976 rpm (fixed)	TRL 4 (GVS-DEC-001 O4)		
S9	The disc brake stops the bowl within the 15 s of PR-012	Physics	Mechanics and statics	t = I omega / (T_brake + T_drag); bicycle disc, mu 0.4, 150 N per pad at 70 mm	0.59 s <= 15 s	Within known science	GVS-CAL-001 section 10
S10	The 6 mm HDPE tub and 10 mm lid contain a fragment of the bowl thrown off at 900 to 1,200 rpm	Physics	Mechanics and statics	Oblique impact: normal share 0.66 of 61 J gives 40 J at 1,200 rpm and 22 J at 900 rpm into the wall; plugging energy of the wall 57 J face first, 14 J edge first (dishing not counted); a quarter of the whole bowl group (review's bounding case) puts 125 J into the wall	Not shown edge first, nor for the bounding case. Fix proposed: a 3 mm steel band (it breaks PR-011, so it goes to Amish, GVS-DEC-001 O2), or an impact test before any spin test	**Fails, fix proposed**	Open: decision for Amish (GVS-DEC-001)
S11	With a full tank the machine stands on a 10 degree slope in any direction	Physics	Mechanics and statics	Tipping about the narrowest base (600 mm) with the frame-borne mass at 0.55 m and a full 64 kg tank on the post	13.1 degrees >= 10 degrees	Within known science	GVS-CAL-001 section 11; GVS-DEC-001 2026-10-03
S12	The spindle members carry the belt pull, rotor weight and unbalance with a safety factor of at least 1.5 on yield	Physics	Strength of materials	Central point load on a pinned 600 mm span of 25 x 25 x 1.5 tube, S235	61 MPa <= 157 MPa	Within known science	GVS-CAL-001 section 11

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S13	The tank post above the braces does not yield at the side push that tips the machine	Physics	Strength of materials	Cantilever bending of 25 x 25 x 1.5 tube above the brace node	112 MPa <= 157 MPa	Within known science	GVS-CAL-001 section 11
S14	The rubber bump stop makes the table stroke asymmetric: the deck stops sharply at the far end and turns smoothly at the head end	Physics	Acoustics and vibration	Impact of the 17 kg deck on a bonded rubber buffer (shape factor), against the harmonic turn at the head end	3.4 deceleration ratio >= 2 deceleration ratio	Within known science	GVS-CAL-001 section 9
S15	The shaking table reduces about 5.3 kg of bowl concentrate to 100 g or less in two passes (53:1)	Physics	Mechanics and statics	Shaking tables reach concentration ratios of 28.8 to 170.9 (mean 66.7) at the MICA plant in Migori, Kenya (Mitchell et al. 2026); a 1,000 x 450 mm table on a heavy-mineral concentrate is not proven	Two passes at 8:1 give 64:1 on paper (PR-005 at risk); test at TRL 4	**Plausible, to confirm at TRL 3**	Open: test or datasheet at TRL 4
S16	The bump stop buffer works within its rating at the largest setting	Engineering and technology	Standards practice	Peak buffer force at the 6 mm setting against the buffer's rated compression load (the BOM said 600 N; now 800 N, as GVS-CAL-001 section 9 chose)	594 N <= 640 N	Within known science	GVS-DEC-001 F3
S17	A 61 rpm cadence turns the bowl at about 730 rpm through 48T/12T, 1:1 and 300/100	Mathematics	Units and dimensional analysis	Product of the stage ratios	claimed 730 rpm, estimate 732 rpm (ratio 1.00)	Within known science	GVS-CAL-001 section 2

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S18	A bicycle computer set to 1,667 mm reads bowl rpm divided by 10	Mathematics	Units and dimensional analysis	Speed = rpm x wheel size x 60 / 1000	claimed 73 km/h at 730 rpm, estimate 73 km/h at 730 rpm (ratio 1.00)	Within known science	GVS-CAL-001 section 2
S19	Water use is 1.19 m3/h: slurry water at 30 % solids plus 12 L/min of fluidization	Mathematics	Orders of magnitude	Mass balance of water	in 1.5 m3/h, out 1.187 m3/h	Within known science	GVS-CAL-001 section 4
S20	The 60 L header drum holds 60 L as drawn	Mathematics	Units and dimensional analysis	Volume of the drawn drum (was 400 mm tall: 45 L)	claimed 60 L, estimate 63.5 L (ratio 0.94)	Within known science	GVS-DEC-001 F1
S21	The gold balance is consistent: 7.75 g in per shift on 5 g/t ore (1.55 t of feed), about 60 % smelted as the product of the assumed stage recoveries (the recoveries themselves are S3 and S15)	Mathematics	Statistics and uncertainty	Product of assumed stage recoveries (0.602); gold in = grade x feed time (was the full 8 h: 8.0 g)	claimed 0.6 fraction, estimate 0.602 fraction (ratio 1.00)	Within known science	GVS-DEC-001 F4; GVS-CAL-001 section 13
S22	Each flush yields about 1.33 kg of concentrate, a mass pull of 0.33 %	Mathematics	Orders of magnitude	Groove volume x 60 % fill x 2.5 kg/L wet bulk	claimed 1.33 kg, estimate 1.33 kg (ratio 1.00)	Within known science	GVS-CAL-001 section 8
S23	With the motor the bowl cannot exceed 900 rpm	Engineering and technology	Control	The 1.2:1 step-up is chosen from an assumed 250 rpm no-load motor speed (no datasheet), so the kinematic cap holds only if the motor's real no-	To confirm against the motor datasheet when bought (GVS-DEC-001, to confirm item 10)	**Plausible, to confirm at TRL 3**	Open: test or datasheet at TRL 4

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				load speed at full pack voltage is 250 rpm or less; MotionCore's own speed limit is the second cap			
S24	The fluidization water is clean enough for 1.0 mm holes: a screen filter stops grains that could block them	Engineering and technology	Standards practice	Fluidization water must be clean (Veiga and Gunson 2020). The 120 mesh screen stops grains over 0.125 mm, an eighth of the hole size; silt finer than that and biofilm still pass, and the right ratio for silty pond water is not sourced here	Flow through every ring after the filter to check at TRL 4 (BLD first checks)	**Plausible, to confirm at TRL 3**	GVS-DEC-001 F2 (filter fitted)
S25	A local workshop builds every part without a lathe; the liner is cast in a printed mold	Engineering and technology	Manufacturing tolerances	Set-screw insert bearings, a taper bush and a welded nipple avoid turning (GVS-DDR-003); casting a Shore 85A PU liner on a six-segment printed core is common practice in PU shops but untried here	Casting route to try at TRL 4 (PR-010)	**Plausible, to confirm at TRL 3**	Open: test or datasheet at TRL 4
S26	A rider supplies 48 W at 60 G for sessions of about an hour, taking turns (the bowl and table never run together)	Biology	Human physiology and biomechanics	Sustained pedalling by an average adult about 75 W (kit limit); at 80 G the 62.8 W is near it (PR-006 at risk)	48.2 W <= 75 W	Within known science	GVS-CAL-001 section 7
S27	Direct smelting of the table concentrate with borax recovers the	Chemistry	Inorganic	Gold melts at 1,064 C; borax fluxes oxide and silicate gangue into a slag. Benguet miners	Depends on the partner's ore; smelting is outside the machine (RSK H10)	**Plausible, to confirm at TRL 3**	Open: test or datasheet at TRL 4

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	gold with no mercury			smelt gravity concentrates with borax (Appel and Na-Oy 2012); the US EPA cites concentrates of about 50 to 100 g. A sulfide-rich (pyrite, arsenopyrite) concentrate can carry gold into a matte and fume arsenic, so it may need the sulfides removed or roasted first			
S28	Mercury in old tailings would be concentrated by the bowl, so the machine must never process mercury-contaminated material	Chemistry	Environmental and atmospheric chemistry	Liquid mercury (13.5 g/cm3) and amalgam are denser than quartz and report to any gravity concentrate	A hazard, not a performance claim: kept as a safety rule in PRC, RSK and the README	Within known science	GVS-RSK-001
S29	The cast PU liner lasts 500 h or more on quartz feed	Chemistry	Materials and polymer chemistry	Elastomer and ceramic wear linings are used in commercial centrifugal concentrator bowls (a moulded rubber insert lasted 1.5 to 2 times the ceramic tiles at Detour Gold; Canadian Mining Journal 2014); life of this cast PU on this feed is unknown	Wear data needed (PR-014, TRL 4)	**Plausible, to confirm at TRL 3**	Open: test or datasheet at TRL 4
S30	Settling ponds and flooded pits near the machine can	Biology	Epidemiology	Malaria transmission is linked to mining sites, where pits	A site risk: pond fencing, covering or draining with	Within known science	GVS-RSK-001

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	breed mosquitoes			and ponds hold standing water (studies of mining workers in Muara Enim, Indonesia, Scientific Reports 2023, and of a malaria outbreak at illegal gold mining sites in French Guiana)	the partner (RSK H7)		
S31	Heavy-mineral concentrates can carry arsenic (arsenopyrite) and lead; gloves and hand washing control the exposure at clean-up (smelter fume is the smelting step's hazard, RSK H10)	Biology	Toxicology	Arsenopyrite is a common gold-bearing sulfide; arsenic is acutely and chronically toxic (WHO arsenic fact sheet)	Safety rule in PRC and RSK	Within known science	GVS-RSK-001
S32	The GFRP shell does not buckle under the full jacket's pressure with the bowl empty and the holes blocked, at 1,200 rpm	Physics	Strength of materials	Windenburg-Trilling short-cylinder buckling of the 4 mm shell between the riffle rings (38 mm pitch), E = 7 GPa hand-laid GFRP, liner ignored (independent review's missing case)	139 kPa <= 2259 kPa	Within known science	GVS-CAL-001 section 10
S33	A full tank on the 1.7 m post does not resonate with the table drive	Physics	Acoustics and vibration	Post above the brace node as a cantilever under the 64 kg tank: about 3.5 Hz, against the table at 4.0 to 5.0 Hz and the harmonics of the bump stop impacts	Near resonance. Fix proposed: raise the brace node to 1.6 m (about +0.5 kg, which takes PR-011 over 100 kg) or run the table with	**Fails, fix proposed**	Open: decision for Amish (GVS-DEC-001)

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				(independent review); with the brace node raised to 1.6 m about 7.5 Hz	the drum drained; to Amish (GVS-DEC-001 O3)		
B1	Option B: the 12 mm diaphragm stroke lifts the water over the ragging bed by at least 5 mm each pulse	Physics	Fluid mechanics	Displaced volume = effective diaphragm area x stroke, spread over the screen area	5.7 mm >= 5 mm	Within known science	GVS-OPT-001 (model: 190 mm plate in a 216 mm free opening)
B2	Option B: the pedal drives the jig with about 15 W	Physics	Thermodynamics	Work to lift the bed pressure over the displaced volume, twice for losses, at 240 strokes/min	claimed 15 W, estimate 8.75 W (ratio 1.71)	Within known science	GVS-OPT-001
B3	Option B holds free gold coarser than about 0.1 mm but loses much of the finer gold	Physics	Fluid mechanics	Jigs are not efficient for fine gold particles and need narrow size fractions (Veiga and Gunson 2020)	Within known science; scored against UR-003 in OPT	Within known science	GVS-OPT-001
B4	Option B uses about 2.3 m3/h of water (hutch water at about 6 mm/s plus slurry water)	Mathematics	Orders of magnitude	Hutch upflow over the bed area plus slurry water; jigs need large amounts of water (Veiga and Gunson 2020)	claimed 2.3 m3/h, estimate 2.36 m3/h (ratio 0.97)	Within known science	GVS-OPT-001
C1	Option C holds gold coarser than about 0.1 mm; recovery falls below 20 % for gold finer than 100 um	Physics	Fluid mechanics	Sluice recovery drops to less than 20 % for grains finer than 100 um (Mitchell et al. 2026); good recovery of 105 to 212 um gold with expanded metal riffles (Poling and Hamilton, UBC pilot study)	Within known science; scored against UR-003 in OPT	Within known science	GVS-OPT-001
C2	Option C uses about 2 m3/h	Mathematics	Orders of magnitude	Water at 10 % solids by mass for	claimed 2 m3/h, estimate 1.8		GVS-OPT-001

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	of water running at about 10 % solids			200 kg/h of solids; large sluices use far more (Veiga and Gunson 2020)	m3/h (ratio 1.11)	Within known science	
C3	Option C's concentrate (about 1 to 2 % of the feed, 15 to 30 kg per shift) is panned by hand to a smeltable 100 g	Biology	Ergonomics	Sluices concentrate weakly (mean ratio 2.5 at Migori; Mitchell et al. 2026); panning tens of kilograms a day is hours of stooped work	Plausible on paper; the labour is scored in OPT	**Plausible, to confirm at TRL 3**	GVS-OPT-001 (not carried forward)

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. The kit's verdict label says "to confirm at TRL 3"; for this TRL 3 repo the open rows can only be closed by a test, which is TRL 4 work and on hold.

Fields not relevant

- Physics, electromagnetism and circuits: GravitySort has no circuit of its own beyond a bicycle computer; the motor option's electrics are MotionCore's and are checked in that repository.
- Physics, optics and radiation; nuclear and ionising radiation: nothing in the design emits or depends on light or radiation.
- Chemistry, electrochemistry and batteries: only through the MotionCore pack (MotionCore repository); galvanic corrosion of the stainless spindle in the painted steel frame is a maintenance point, not a performance claim.
- Chemistry, organic and analytical: no reagents beyond water; borax is used at the smelting step outside the machine (S27).
- Biology, microbiology, plant and animal biology, biochemistry: no living material in the process; mosquitoes in ponds are covered under epidemiology (S30).

Concept options

- **Option A, fluidized bowl and shaking table (current design):** no row fails outright; S10 (containment) and S33 (tank sway) fail with fixes that go to Amish; plausible rows S3, S8, S15, S23, S24, S25, S27 and S29 need tests or datasheets.
- **Option B, diaphragm jig and shaking table:** no row fails (B1 to B4). A first model of the jig with a 124 mm diaphragm gave only about 2 mm of water stroke over the bed, below what pulses the ragging; the option was modelled with a 190 mm plate in a 216 mm opening (5.7 mm) before it was scored. Jigs lose much of the gold finer than about 0.1 mm (B3), which is scored against UR-003 in GVS-OPT-001.
- **Option C, carpeted sluice and gold pans:** no row fails (C1 to C3); sluices recover less than 20 % of gold finer than 100 um (C1), which is scored against UR-003, and the panning labour is plausible on paper only (C3).

Independent review

A second agent (Claude, run headless with read access to four files only: this STEM check with `stem_check.py`, GVS-CAL-001, GVS-PRS-001 and `sizing.py`, and none of the author's reasoning) was asked on 2026-10-25 to refute the calculation note's headline results and every "Within known science" row. It could not run the script (it needs the CAD model) and worked by hand. Its findings and what changed:

#	Finding	Verdict given	What changed
1	The fluidization supply ignored the 0.46 to 0.57 m climb from the union to the rings, 4.5 to 5.6 kPa; the 0.5 kPa margin of S4 disappears	Wrong	Accepted. Climb added to every ring (`ring_dp`); the lowest ring fell to 6.0 kPa against the proposed 10 kPa, so S4 failed. Fixed under the standing instruction (F6): 3/4 in valve and rotameter cut the fittings loss from 7.5 to 3.4 kPa; 10.0 kPa net at mid-tank, 7.0 kPa in the worst case. No pump, as Amish decided on 2026-10-02
2	The 2.7 mm/s upflow is below the minimum fluidization velocity (Wen and Yu) of particles over about 75 um at 60 G, so the bed is not fully fluidized and the 80 % recovery of 75 to 1,000 um gold is unsupported; S3 should be Fails	Plausible should be Fails	Partly accepted. The Wen and Yu check is now in the CAL (2.9 mm/s for 75 um quartz, 27 mm/s for 150 um pyrite) and in S3. S3 stays Plausible because commercial fluidized bowls also run below full fluidization of the coarse bed and their recovery is empirical; flow and pressure are at the low end of commercial practice, so the TRL 4 recovery test must sweep the fluidization flow (GVS-DEC-001 O5)
3	The critical speed assumed rigid bearings; the frame members are as compliant as the shaft, so it may be 1,400 to 1,650 rpm, under 1.25 x 1,200 rpm	Weak	Accepted. Support compliance added: 1,274 to 1,976 rpm (member ends pinned to fixed). S8 is now Plausible, to confirm by a tap test on the first frame; still well above the 900 rpm limit (GVS-DEC-001 O4)
4	Missing case: shell buckling under the full jacket with the bowl empty and the holes blocked	Missing	Accepted. New row S32: 139 kPa outside against 6.8 MPa with the riffle rings as stiffeners, safety factor 49
5	Pedal power: bowl and table together would need 63 W	Stands, low confidence	No change: the bowl and the table never run together (the table runs at clean-up with the bowl stopped); stated in S6 and S26
6	Braking stands; the lid lock follows the brake cable, not zero speed	Stands	No change: the interlock needs the brake parked, which stops the bowl in 0.6 s
7	Containment: the quarter of liner and shell is arbitrary; a quarter of the whole bowl group puts about 105 J into the wall; the 12 mm edge contact is questionable; call it Fails until tested	Weak, Fails	Accepted. Bounding case added (125 J at 1,200 rpm). S10 is now Fails, fix proposed: the steel band breaks PR-011, so the choice goes to Amish (O2)
8	Tipping stands; the post with a full tank sways at about 3.5 Hz, inside the 4 to 5 Hz table band	Stands with caveats	Accepted. New row S33, Fails, fix proposed: raising the brace node to 1.6 m gives about 7.5 Hz but adds about 0.5 kg over PR-011, so it goes to Amish (O3)
9	Mass margin of 0.4 kg on an estimate: mark R11 at risk	Weak	Accepted: PR-011 is now at risk
10	Water use omits table wash water and the lift	Weak	No change to PR-008: the table runs at clean-up, not with the bowl; the lift is the site rule decided by Amish on 2026-10-02 (gravity from upstream or a pump at the site)
11	Table concentration ratios at Migori are grade ratios, not mass reductions	Weak	Accepted; S15 wording keeps it Plausible and says so
12	S21 multiplies assumed recoveries	Circular	

#	Finding	Verdict given	What changed
			Accepted: S21 now claims only the arithmetic consistency of the balance; the recoveries are S3 and S15
13	S6 and S23 are tautologies	Wrong	Accepted. S6 now checks the slurry power term against m v^2 at the lip; S23 is Plausible until the motor datasheet confirms its no-load speed
14	S16 passes because the BOM was edited; static rating against an impact load	Weak	Kept as Within (the CAL had chosen 800 N); the buffer is on the list to confirm when bought (GVS-DEC-001 item 11)
15	S24 restates the design choice; the one-third rule is not sourced	Weak	Accepted: S24 is now Plausible, confirmed by the flow check after the filter at TRL 4
16	S27: borax smelting of a sulfide-rich, arsenic-bearing concentrate loses gold to matte and fumes arsenic	Weak	Accepted: S27 is now Plausible; RSK H10 notes sulfide removal or roasting may be needed
17	S31: hand washing does not control smelter fume	Weak	Accepted: S31 now covers clean-up only; fume is RSK H10
18	S2: the 5 mm film is assumed (a laminar film would be about 0.2 mm) and hindered settling cuts velocities to about a quarter; the ratio of 36 becomes about 9	Stands	No change; still passes
19	B1's 5 mm threshold is unsourced and the plate was enlarged until it passed	Weak	Noted; B is not carried forward unless Amish chooses it
20	S1, S11, S12, S13, S14, S17 to S20, S22, S28, S30, B2 to B4, C1, C2	Stand	No change

Sources

- Appel, Peter W. U., and Leoncio Na-Oy. "The Borax Method of Gold Extraction for Small-Scale Miners." *Journal of Health and Pollution* 2, no. 3 (2012). https://www.journalhealthpollution.org/doi/full/10.5696/2156-9614-2.3.5.
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